VLSDA: Vision–Language Model-Supervised Domain Adaptation for Cross-Domain Object Detection in Remote Sensing

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***Abstract*—Cross-domain object detection in remote sensing suffers from substantial domain gaps arising from differences in resolution, viewing geometry, and imaging modality across sensors and platforms. Existing unsupervised domain-adaptive object detection (DAOD) methods typically align source and target features using the detector's own target-domain representations. However, the extraction of these representations is constrained by the very domain discrepancies they aim to bridge, resulting in noisy and biased features that make alignment unstable. To address this limitation, we propose the vision–language model-supervised domain adaptor (VLSDA), a domain adaptation framework supervised by a frozen vision–language model (VLM). It leverages a frozen VLM image encoder as an additional and stable semantic domain to guide domain alignment. Our VLM-supervised prototypical alignment (VLPA) module stabilizes categorywise alignment through a tridomain adversarial strategy that jointly aligns source–VLM, target–VLM, and source–target distributions. Complementing this, the global cross-domain contrastive alignment (GCCA) module enhances intraclass compactness and interclass separability via supervised contrastive learning. Without requiring any fine-tuning of the VLM, our framework directly mitigates reliance on noisy target features and improves robustness to large distribution shifts. Extensive experiments on multiple cross-domain remote sensing benchmarks demonstrate consistent improvements over state-of-the-art methods, including 68.3% mAP_{50} on xView → DOTA and 70.5% on High-Resolution Remote Sensing Detection (HRRSD) dataset → SAR Ship Detection Dataset (SSDD). The code is available at: https://github.com/JunhongLu0704/VLSDA**

***Index Terms*—Cross-domain alignment, domain-adaptive object detection (DAOD), remote sensing, vision–language model (VLM).**

I. Introduction

REMOTE sensing object detection (RSOD) plays a crucial role in disaster monitoring, land resource management, and urban planning [1], [2], [3]. Recent advances in deep learning have further improved model structures for RSOD, achieving strong results even in challenging scenarios [4]. However, remote sensing images collected from various sensors and platforms vary markedly in resolution, viewing geometry, and imaging modality. These variations distort foreground–background statistics, creating severe domain gaps, so detectors trained on one domain often generalize poorly to another. Meanwhile, acquiring high-quality annotations for every domain is prohibitively expensive. Hence, unsupervised domain-adaptive object detection (DAOD) has emerged as a practical solution: it learns from labeled source data and unlabeled target data to achieve robust performance on the target domain despite distribution shifts.

Most existing domain adaptation methods aim to align feature distributions between source and target domains to mitigate distribution shifts. In object detection, this problem is harder because detectors must both localize objects and separate object features from background, which implies identifying reliable object features in the unlabeled target domain. However, a detector trained on source data often performs poorly when applied to the unlabeled target domain due to the very distribution shifts that DAOD seeks to mitigate; consequently, the target-side features it extracts are noisy and often fail to disambiguate object features from background. For example, a source-domain object feature can be mistakenly aligned with a background feature in the target domain. Such misalignment amplifies, rather than resolves, cross-domain discrepancies and yields suboptimal adaptation. Thus, methods that rely primarily on source–target feature alignment are constrained because their alignment hinges on target-domain features estimated by a source-trained detector, which are themselves corrupted by the shift. This observation indicates that relying on the detector alone is unlikely to suffice for robust cross-domain adaptation. Recent advances [5] demonstrate that external guidance can enhance the generalization of task-specific models. Vision–language models (VLMs), pretrained on massive and diverse image–text pairs, learn semantically consistent visual representations across heterogeneous contexts, endowing them with strong cross-domain generalization. This makes VLMs a natural candidate for external semantic reference in domain adaptation. However, existing efforts to exploit VLMs for detection are not directly compatible with DAOD. Open-vocabulary detection methods extend detectors with large text encoders and intensive vision–language fusion to recognize unseen categories. In contrast, DAOD does not involve unseen categories—its challenge lies in distribution shifts where target images are already available. Expanding a textual branch thus increases

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model complexity without improving the efficiency of adaptation. Distillation-based approaches, such as VLDadaptor [6], maintain conventional source–target alignment and inject VLM-derived knowledge as an auxiliary constraint. However, as the VLM remains external to the alignment process, adaptation still hinges on the coupling between unstable target features and the source domain—making the supervision susceptible to noise and limiting robustness under large domain gaps.

To address these limitations, we reinterpret the role of the VLM from an external supervisor to an equivalent third domain that complements the source–target pair. By incorporating the frozen VLM image encoder as an additional and comparatively stable semantic space, we form a triangular alignment among the source, target, and VLM domains. Unlike conventional two-domain alignment that enforces direct coupling between source and target features, our design introduces mutual constraints among all three domains. During optimization, each domain is simultaneously attracted toward the other two, forming a geometric equilibrium where representations converge to the shared semantic centroid rather than drifting toward any single domain. Unlike distillation-based methods, where supervision remains external to alignment, our design embeds the VLM as a peer domain, transforming alignment from pairwise coupling into triangular convergence.

Building on this idea, we propose that the VLM-supervised domain adaptor (VLSDA) leverages a frozen VLM image encoder as an additional and stable semantic domain to guide domain alignment in DAOD. It consists of two components: VLM-supervised prototypical alignment (VLPA) and global cross-domain contrastive alignment (GCCA).

VLPA stabilizes prototypewise feature alignment through a tridomain adversarial mechanism. Unlike conventional dual-domain setups that depend solely on noisy source–target interactions, VLPA introduces three discriminators to align the distributions of source–target, source–VLM, and target–VLM pairs simultaneously. This triangular alignment imposes mutual consistency among all domains, preventing any single domain (especially the target) from dominating the optimization. Consequently, source and target features are progressively attracted toward domain-agnostic VLM anchors, forming stable semantic prototypes that encapsulate cross-domain object semantics. These prototypes serve as both a regularizing reference and a transferable foundation for subsequent contrastive refinement.

GCCA complements VLPA by explicitly reinforcing intraclass compactness and interclass separation across domains. It maintains a global memory of categorywise prototypes drawn jointly from the detector features and frozen VLM embeddings. By applying supervised contrastive learning where category identities serve as supervision signals, GCCA strengthens the semantic topology of the shared feature space, ensuring that domain-invariant prototypes remain discriminative. This refinement further enhances the quality of the prototypes that VLPA relies on, completing a mutually reinforcing loop between adversarial alignment and semantic contrast.

The main contributions are summarized as follows.

1) We propose VLSDA, a novel domain adaptation framework supervised by a frozen VLM image encoder that serves as an additional and stable semantic domain to guide cross-domain alignment, reducing reliance on unstable target-domain representations.
2) We design two complementary modules to jointly improve alignment stability and feature discriminability. VLPA employs tridomain adversarial learning to align feature distributions with VLM semantics, while GCCA uses supervised contrastive learning to structure the cross-domain feature space.
3) Extensive experiments on four cross-domain remote sensing benchmarks demonstrate the effectiveness and generalizability of VLSDA. We achieve 68.3%, 79.8%, 79.9%, and 70.5% mAP_{50} on xView → DOTA, UCAS-AOD → CARPK, CARPK → UCAS-AOD, and High-Resolution Remote Sensing Detection (HRRSD) dataset → SAR Ship Detection Dataset (SSDD), consistently surpassing existing state-of-the-art methods.

II. Related Works

A. Domain-Adaptive Object Detection

Recent studies have achieved remarkable progress in RSOD. SOD-YOLOv10 [7] improves the YOLO framework by enhancing the backbone design and improving the loss of optimization for small object detection. BAFNet [8] proposes a dual-stream attention mechanism combined with boundary-aware fusion. OriMamba [9] incorporates a hybrid Mamba pyramid and a dynamic double head for oriented detection. MaDiNet [4] employs gamma diffusion and Mamba-based context modeling for synthetic aperture radar (SAR) object detection.

Despite their success, these methods assume identical training and testing domains, leaving the cross-domain generalization problem unresolved; this gap is exactly what DAOD aims to fill. DAOD aims to mitigate the performance degradation caused by domain shifts between a labeled source domain and an unlabeled target domain.

Among existing DAOD approaches, feature alignment has emerged as the most widely adopted strategy. They often utilize adversarial training based on gradient reversal layers (GRLs) [10] to learn domain-invariant feature mappings. Early work focused on aligning global features [11]. The following work emphasized local region representations by incorporating region proposals to facilitate finer grained alignment [12]. KTNet [13] extends the alignment to the category level by modeling classwise prototypes. PARPN [14] incorporates prototype alignment into the region proposal network. SIGMA [15] and SIGMA++ [16] introduce a bidirectional graph matching strategy to align category-level distributions. In the remote sensing field, FADA [17] introduces adversarial foreground alignment and prototype-based confusing feature alignment in remote sensing. Remote Sensing Teacher [18] employs frequency-enhanced attention for domain adaptation, while DATR [19] introduces classwise prototypical alignment and dataset-level contrastive learning for fine-grained consistency. FIE-Net [20] enhances the emphasis on foreground

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features and balances the disparity between background and foreground.

Other approaches have also been explored in DAOD. Style transfer-based methods adopt GANs such as CycleGAN [21] to translate source images into the target style with inherited annotations, enabling detectors to be trained as if operating in the target domain [22], [23]. Self-training methods, often based on the mean teacher framework [24], refine detectors on unlabeled target data through iterative pseudo-labeling, with extensions such as CMT [25] integrating contrastive learning for robustness and MTTrans [26] combining Transformers with multilevel feature alignment. More recently, Transformer-based detectors, such as DETR [27], reformulate detection as a direct set prediction task with global self-attention. Variants including SFA [28], which introduces domain-query and tokenwise alignment, and AQT [29], which incorporates adversarial token and cross-attention modules, enhance cross-domain consistency in DETR. Although diverse in form, these methods still largely depend on feature alignment.

Despite their progress, existing DAOD approaches remain confined to binary alignment between the source and target domains. Under large distribution shifts, such two-domain alignment often struggles to extract reliable target-domain object features, leading to misaligned representations and degraded performance.

B. VLMs for Detection and Adaptation

VLMs have demonstrated remarkable cross-modal generalization by pretraining on hundreds of millions of image–text pairs, such as CLIP [30] and ALIGN [31]. These models learn a shared embedding space in which visual and textual representations are directly comparable, enabling powerful semantic alignment across modalities. In remote sensing, RemoteCLIP [32] unifies heterogeneous annotation schemes, and GeoRSCLIP [33] further leverages the RS5M corpus to fine-tune CLIP's on geospatial image–text pairs, achieving superior zero-shot performance on downstream RS benchmarks.

Beyond classification, VLMs have been extended to more complex downstream tasks. RegionCLIP [34] and UniDetector [35] explore open-vocabulary detection by aligning region-level representations with text. Grounding DINO [36] integrates full-phase multimodal fusion for strong referring detection. OpenRSD [37] and LAE-DINO [38] adapt VLMs into remote sensing open-vocabulary detection. VLDadaptor [6] distills VLM knowledge into DAOD, introducing domain-mixed contrastive and consistency distillation. ProFiT [5] further demonstrates the value of Transformer-based token interaction and progressive external guidance in enhancing model generalization.

Despite recent advances, how VLMs should be incorporated into remote sensing DAOD remains unclear. Open-vocabulary detectors rely on heavy text encoders and cross-modal token interactions, increasing parameter overhead while focusing on vocabulary expansion rather than stabilizing cross-domain alignment. Distillation-based approaches keep VLM supervision external to the alignment mechanism, leaving optimization dominated by binary source–target coupling and hence sensitive to unstable target features under large shifts. These limitations call for rethinking the VLM's role beyond auxiliary supervision or language-heavy designs.

III. Proposed Method

To address the instability of feature alignment under large domain shifts in RSOD, we reinterpret the VLM as an additional and comparatively stable semantic domain on par with the source and target. Concretely, we incorporate a frozen VLM image encoder and formulate a triangular alignment among the source, target, and VLM domains, so that features naturally balance toward domain-invariant semantics rather than being forcibly pulled or dominated by noisy target References. Our instantiation comprises two complementary modules: VLPA, which performs instance-level prototype alignment to steer S/T features toward the VLM semantic anchor, and GCCA, which enforces global class-conditional consistency to maintain discriminability after alignment. Notably, our framework requires no text encoder and no VLM fine-tuning, keeping computational and parameter overhead minimal.

In the following subsections, we first define the domain adaptation problem setup in Section III-A, then describe prototype construction in Section III-B, followed by detailed explanations of the VLPA and GCCA modules in Sections III-C and III-D, respectively, and finally present the network training strategy in Section III-E.

A. Overview

1) Problem Formulation: We consider the standard unsupervised DAOD setting, where a labeled source domain $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ and an unlabeled target domain $\mathcal{D}_t = \{x_j^t\}_{j=1}^{N_t}$ are provided. Here, x_i^s and x_j^t denote input images, and y_i^s represents the corresponding bounding-box coordinates and category labels. Both domains share the same set of object categories but exhibit significant appearance discrepancies due to differences in resolution, viewing geometry, and imaging modality. The objective is to train a detector that generalizes well on $\mathcal{D}_t$ despite significant domain shifts.

2) Framework: VLSDA introduces a frozen VLM image encoder as a stable semantic anchor to stabilize category-level alignment under domain shifts. The VLM, pretrained on large-scale image–text pairs, provides comparatively domain-agnostic, class-consistent features. Rather than fine-tuning or incorporating heavy text encoders as in open-vocabulary detection, VLSDA adopts a lightweight design that keeps the VLM frozen and extracts semantic prototypes directly from cropped object regions, offering stable external references for feature alignment.

As illustrated in Fig. 1, the framework integrates two complementary alignment modules. VLPA focuses on stabilizing intraclass feature aggregation by employing tridomain adversarial learning that simultaneously aligns source, target, and VLM prototypes for each category. This triangulated alignment pulls same-class features across domains toward shared semantic centers, mitigating domain bias even under class imbalance. In parallel, GCCA imposes a global classwise

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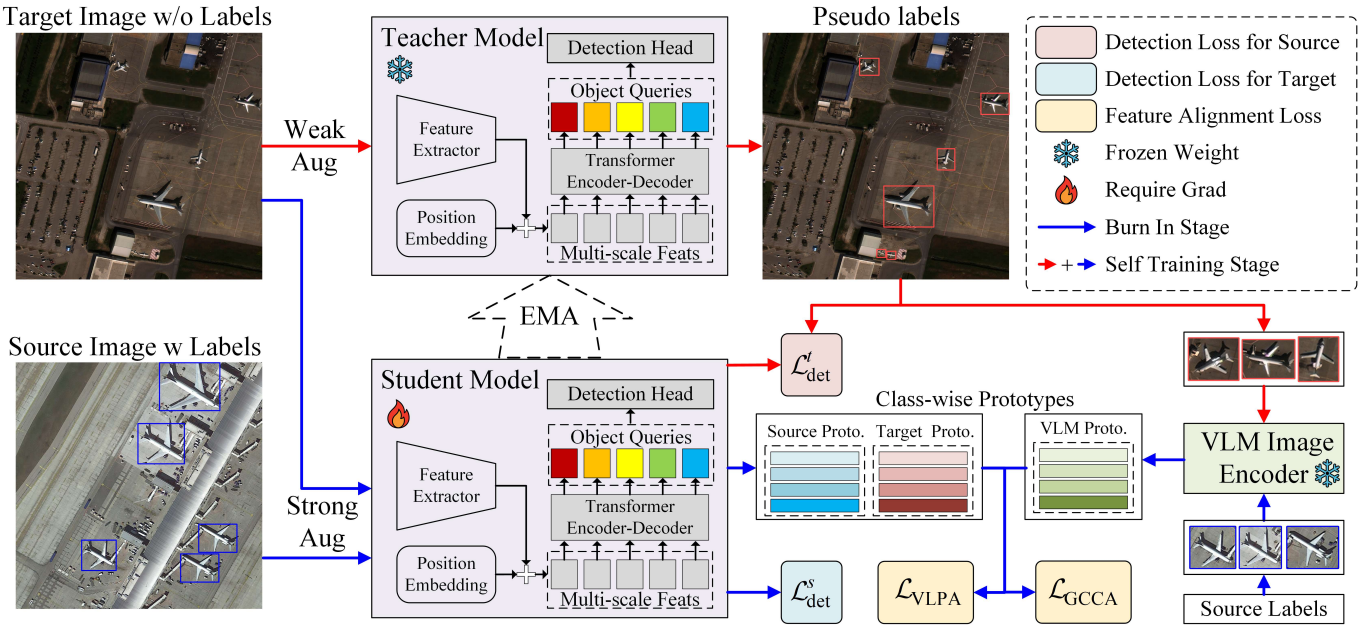


Fig. 1. Overview of the proposed VLSDA framework. VLSDA operates under a teacher–student paradigm built upon a DINO-based detector. During training, the source images with labels and the target images without labels are fed into the student model. The teacher model generates pseudo-labels for the target domain, which are used both for detection supervision and to crop object regions for constructing VLM prototypes via the frozen vision–language encoder. VLPA performs intraclass tridomain adversarial alignment among source, target, and VLM prototypes. GCCA conducts classwise-supervised contrastive learning using global semantic centers accumulated from DINO and VLM features.

structure by applying supervised contrastive learning over category prototypes accumulated from both DINO features and VLM embeddings, simultaneously enhancing intraclass compactness and interclass separability across domains. Together, they form a closed-loop adaptation: VLPA ensures "where to align," while GCCA ensures "how to separate."

VLSDA operates under a teacher–student paradigm: the teacher generates pseudo-labels for the target domain, which not only provide additional supervision to the student detector but also serve as object region proposals for constructing VLM prototypes. Both VLPA and GCCA modules are jointly optimized throughout training under this unified framework.

B. Prototype Construction

VLSDA constructs two sets of categorywise prototypes to provide semantic anchors for cross-domain alignment: 1) external prototypes extracted from the frozen VLM and 2) internal prototypes derived from the DINO-based detector. These prototypes serve as inputs for the VLPA and GCCA modules, but are utilized with distinct alignment objectives in each case.

1) VLM Prototype Construction: Given an input image, object-level regions are first cropped based on detection boxes. In the source domain, these boxes are directly from ground-truth annotations. In the self-training stage, pseudo-labels predicted by the teacher provide additional bounding boxes for the target domain.

Let $\mathbf{r}_n$ denote the nth cropped object region, which is passed through the frozen VLM image encoder $\mathcal{F}_{\text{VLM}}(\cdot)$ to obtain a semantic embedding

$$\mathbf{f}_n^{\text{VLM}} = \mathcal{F}_{\text{VLM}}(\mathbf{r}_n), \quad \mathbf{f}_n^{\text{VLM}} \in \mathbb{R}^{d_{\text{VLM}}} \tag{1}$$

where d_{VLM} denotes the VLM embedding dimension.

For each class $k \in \mathcal{C}$, where $\mathcal{C}$ is the predefined category set, we aggregate the embeddings of all object instances assigned to the class k. Denoting $\mathcal{I}_k$ as the index set of such instances, the categorywise VLM prototype is computed as

$$\mathbf{p}_k^{\text{VLM}} = \frac{1}{|\mathcal{I}_k|} \sum_{n \in \mathcal{I}_k} \mathbf{f}_n^{\text{VLM}}. \tag{2}$$

2) DINO Prototype Construction: In parallel, internal prototypes are extracted from the DINO-based detector. Let $\{\mathbf{q}_i\}_{i=1}^{N_q}$ be the object queries output by the last DINO decoder, where $N_q = 900$ is the number of decoder slots following DINO [39]. Each query $\mathbf{q}_i \in \mathbb{R}^{d_{\text{DINO}}}$ is assigned a predicted class via

$$\hat{y}_i = \arg\max_k \sigma(f_{\text{cls}}(\mathbf{q}_i)) \tag{3}$$

where $f_{\text{cls}}(\cdot)$ is the detector's classification head and $\sigma(\cdot)$ denotes the sigmoid activation.

For each class k, we collect the set $\mathcal{J}_k = \{i \mid \hat{y}_i = k\}$ and compute the classwise DINO prototype

$$\bar{\mathbf{p}}_k^{\text{DINO}} = \frac{1}{|\mathcal{J}_k|} \sum_{i \in \mathcal{J}_k} \mathbf{q}_i. \tag{4}$$

As DINO and VLM embeddings reside in different feature spaces, a projection head $\phi : \mathbb{R}^{d_{\text{DINO}}} \to \mathbb{R}^{d_{\text{VLM}}}$ is applied to map DINO prototypes into the VLM space

$$\mathbf{p}_k^{\text{DINO}} = \phi\left(\bar{\mathbf{p}}_k^{\text{DINO}}\right). \tag{5}$$

The resulting prototypes $\{\mathbf{p}_k^{\text{DINO}}\}$ and $\{\mathbf{p}_k^{\text{VLM}}\}$ are used in the downstream modules. VLPA employs these prototypes to conduct tridomain adversarial learning, pulling source, target, and VLM features within each category toward shared

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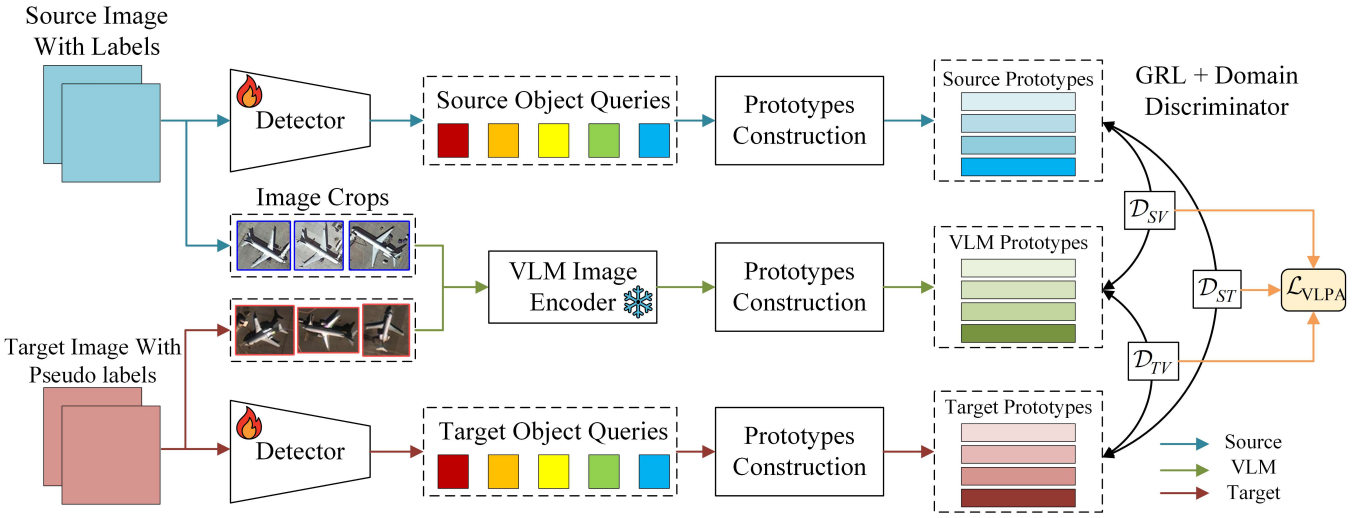


Fig. 2. Architecture of the proposed VLPA module. Object queries from both source and target domains are aggregated into categorywise prototypes. VLM prototypes are obtained via frozen vision–language encoders on object crops. Tridomain adversarial alignment is conducted through domain discriminators $\mathcal{D}_{\mathrm{SV}}, \mathcal{D}_{\mathrm{ST}}$, and $\mathcal{D}_{\mathrm{TV}}$, supervised via GRL, encouraging domain-invariant semantic feature alignment.

semantic centers. GCCA accumulates DINO and VLM prototypes across batches into global memory queues, enforcing supervised contrastive learning to simultaneously optimize intraclass compactness and interclass separability.

C. VLM-Supervised Prototypical Alignment

Prior domain-adaptive detectors often align source and target features directly through mutual attraction in the embedding space. However, this strategy may suffer from class imbalance and noisy supervision. As shown in GPA [40], such alignment tends to collapse toward dominant classes, while failing to represent multimodal category distributions within each class. Similarly, ACIA [41] highlights that class-agnostic adversarial alignment without explicit semantic guidance is vulnerable to category-level confusion under large domain shifts.

To address these limitations, VLPA introduces an additional and stable semantic domain derived from frozen VLM prototypes and adopts tridomain adversarial learning, which aggregates prototypes from source, target, and VLM domains within each category. This stabilizes feature aggregation under severe domain discrepancies by providing category-aware alignment constraints.

For each class $k \in \mathcal{C}$, we consider three prototypes: source-domain DINO prototype $\mathbf{p}_k^{\mathrm{DINO}_s}$, target-domain DINO prototype $\mathbf{p}_k^{\mathrm{DINO}_t}$, and external VLM prototype $\mathbf{p}_k^{\mathrm{VLM}}$. These prototypes represent class-level feature centers in their respective domains.

Instead of directly minimizing distances between all prototypes, we adopt adversarial learning to implicitly aggregate these features. Fig. 2 illustrates the workflow of the proposed VLPA module. For each category k, three prototypes are passed to independent domain discriminators $\mathcal{D}_{\mathrm{SV}}, \mathcal{D}_{\mathrm{ST}}$, and $\mathcal{D}_{\mathrm{TV}}$ Each discriminator is responsible for distinguishing one domain pair: $\mathcal{D}_{\mathrm{SV}}$ aligns source and VLM prototypes, $\mathcal{D}_{\mathrm{ST}}$ aligns source and target prototypes, and $\mathcal{D}_{\mathrm{TV}}$ aligns target and VLM prototypes.

Each discriminator is implemented as a lightweight three-layer MLP. Before each discriminator, a GRL is inserted to enable adversarial training. During the forward pass, the GRL behaves as an identity mapping, allowing features from the detector to be passed unchanged to the domain discriminator. However, during the backward pass, it multiplies the gradients flowing back from the discriminator by a negative scalar. As a result, the discriminator is trained to distinguish domain-specific prototypes, while the detector backbone is adversarially trained to produce domain-agnostic features that confuse the discriminator. The adversarial losses are defined as follows:

$$\mathcal{L}_{\mathrm{adv}}^{\mathrm{SV}} = \frac{-1}{|\mathcal{C}_b|} \sum_{k \in \mathcal{C}_b} \left[\log \mathcal{D}_{\mathrm{SV}} \left(\mathbf{p}_k^{\mathrm{DINO}_s} \right) + \log \left(1 - \mathcal{D}_{\mathrm{SV}} \left(\mathbf{p}_k^{\mathrm{VLM}} \right) \right) \right] \tag{6}$$

$$\mathcal{L}_{\mathrm{adv}}^{\mathrm{ST}} = \frac{-1}{|\mathcal{C}_b|} \sum_{k \in \mathcal{C}_b} \left[\log \mathcal{D}_{\mathrm{ST}} \left(\mathbf{p}_k^{\mathrm{DINO}_s} \right) + \log \left(1 - \mathcal{D}_{\mathrm{ST}} \left(\mathbf{p}_k^{\mathrm{DINO}_t} \right) \right) \right] \tag{7}$$

$$\mathcal{L}_{\mathrm{adv}}^{\mathrm{TV}} = \frac{-1}{|\mathcal{C}_b|} \sum_{k \in \mathcal{C}_b} \left[\log \mathcal{D}_{\mathrm{TV}} \left(\mathbf{p}_k^{\mathrm{DINO}_t} \right) + \log \left(1 - \mathcal{D}_{\mathrm{TV}} \left(\mathbf{p}_k^{\mathrm{VLM}} \right) \right) \right] \tag{8}$$

where $\mathcal{C}_b \subset \mathcal{C}$ denotes the set of categories present in the current mini-batch. The total VLPA loss aggregates the three domain pairwise objectives

$$\mathcal{L}_{\mathrm{VLPA}} = \frac{1}{3} \left(\mathcal{L}_{\mathrm{adv}}^{\mathrm{SV}} + \mathcal{L}_{\mathrm{adv}}^{\mathrm{ST}} + \mathcal{L}_{\mathrm{adv}}^{\mathrm{TV}} \right). \tag{9}$$

By adversarially confusing domain discriminators across these three domain pairs, VLPA explicitly promotes intraclass aggregation among source, target, and VLM features, stabilizing semantic alignment under large cross-domain discrepancies.

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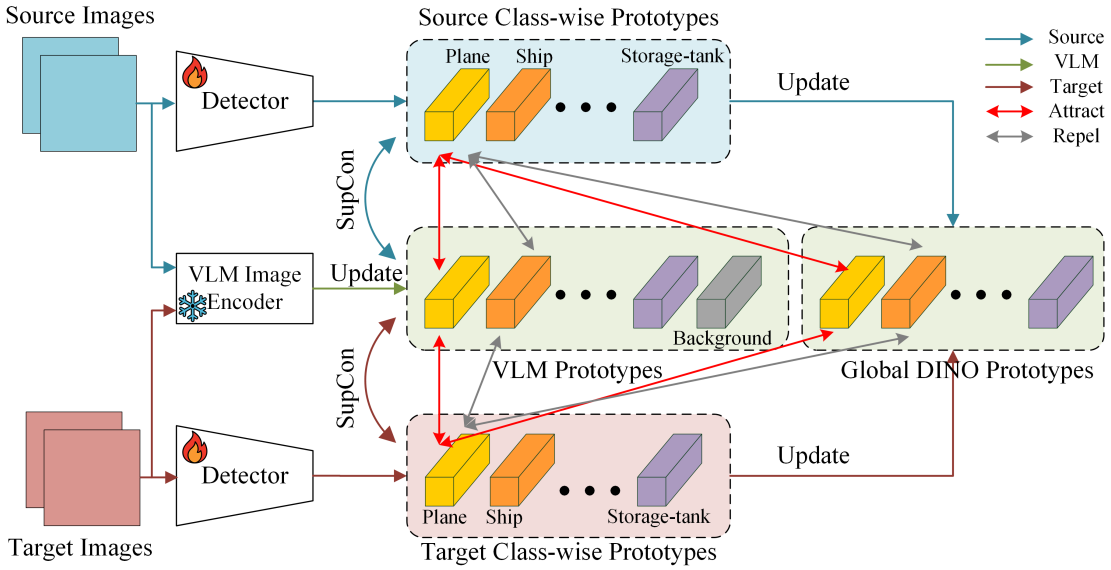


Fig. 3. Architecture of GCCA. Source and target object queries are aggregated into classwise prototypes, while VLM prototypes are extracted via a frozen vision–language encoder. Global DINO prototypes are updated cumulatively. Supervised contrastive learning pulls the same-class prototypes closer while pushing different classes apart across domains.

D. Global Cross-Domain Contrastive Alignment

While VLPA aggregates features across domains within each category, it does not explicitly enforce discriminative interclass separation. To further structure the feature space, we introduce GCCA, which simultaneously enhances intraclass compactness and interclass separability across source, target, and VLM domains.

As illustrated in Fig. 3, for each class $k \in \mathcal{C}$, we maintain two global queues: a DINO-based prototype queue $\mathbf{p}_k^{G_{\text{DINO}}}$ and a VLM-based prototype queue $\mathbf{p}_k^{G_{\text{VLM}}}$. These queues accumulate historical prototypes across batches, providing stable global semantic anchors. Similarly, the frozen VLM encoder provides a batchwise external semantic prototype $\mathbf{p}_k^{\text{VLM}}$ by averaging the embeddings of object crops.

To construct temporally stable semantic anchors, we maintain two global memory banks per class: one for DINO-derived representations, $\mathbf{p}_k^{G_{\text{DINO}}}$, and one for VLM-derived representations, $\mathbf{p}_k^{G_{\text{VLM}}}$. These are updated using a cumulative averaging strategy that weights recent prototypes according to their sample count. Specifically, given the number of valid class-k samples in the current mini-batch, n_k^{cur}, and the accumulated count n_k^{prev} from earlier batches

$$\mathbf{p}_k^G \leftarrow (1-\alpha_k)\,\mathbf{p}_k^G + \alpha_k \mathbf{p}_k^{\text{batch}}, \alpha_k = \frac{n_k^{\text{cur}}}{n_k^{\text{cur}} + n_k^{\text{prev}}} \tag{10}$$

where $\mathbf{p}_k^{\text{batch}}$ is the class-k prototype computed from the current batch, which may refer to either $\mathbf{p}_k^{\text{DINO}_{s/t}}$ or $\mathbf{p}_k^{\text{VLM}}$. This moving average update prevents instability due to noisy or rare categories while ensuring adaptability to the most recent feature distributions. The cumulative count is updated as

$$n_k^{\text{prev}} \leftarrow n_k^{\text{prev}} + n_k^{\text{cur}}. \tag{11}$$

In addition to class-specific positive prototypes, we incorporate background representations as structured hard negatives to enhance interclass separability. These background embeddings are extracted from adversarially jittered bounding boxes around detected objects, ensuring minimal foreground overlap. This design introduces semantically disjoint yet spatially relevant negatives, which are especially valuable for improving discriminative feature learning in cluttered remote sensing environments.

Following [42], we adopt a supervised contrastive loss to align DINO prototypes with both global DINO and VLM anchors. For each class k present in the current batch, we define the source and target domain prototypes $\mathbf{p}_k^{\text{DINO}_s}$ and $\mathbf{p}_k^{\text{DINO}_t}$ as queries. We treat their corresponding global prototypes $\mathbf{p}_k^{G_{\text{VLM}}}$ and $\mathbf{p}_k^{G_{\text{DINO}}}$ as positive anchors, while using anchors from all other categories as negatives. To jointly enforce intradomain consistency and interdomain semantic alignment, we design a contrastive loss over prototype pairs with shared semantics

$$\mathcal{L}_{\text{GCCA}} = -\frac{1}{|\mathcal{C}_b|}\sum_{k\in\mathcal{C}_b}\left[\log\frac{\exp\left(\mathbf{p}_k^{\text{DINO}_s}\cdot\mathbf{p}_k^{G_{\text{VLM}}}/\tau\right)}{\sum_{j\in\mathcal{C}}\exp\left(\mathbf{p}_k^{\text{DINO}_s}\cdot\mathbf{p}_j^{G_{\text{VLM}}}/\tau\right)} + \log\frac{\exp\left(\mathbf{p}_k^{\text{DINO}_t}\cdot\mathbf{p}_k^{G_{\text{DINO}}}/\tau\right)}{\sum_{j\in\mathcal{C}}\exp\left(\mathbf{p}_k^{\text{DINO}_t}\cdot\mathbf{p}_j^{G_{\text{DINO}}}/\tau\right)}\right] \tag{12}$$

where τ is a temperature hyperparameter and $\mathcal{C}_b \subset \mathcal{C}$ represents the set of classes in the current batch. The dot product $\mathbf{a}\cdot\mathbf{b}$ denotes the cosine similarity, as all prototypes are ℓ_2-normalized prior to computation. By treating each domain-specific prototype as an individual query, GCCA imposes pairwise constraints that are semantically meaningful and batch robust. Meanwhile, the use of both DINO and VLM global anchors enables alignment with internal consistency and external semantics.

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TABLE I
DATASET OVERVIEW FOR DIFFERENT DOMAIN ADAPTATION SCENARIOS

Domain Adaptation Scenario	Dataset	Sensor	Class	GSD	Image Size	Images	Annotations
between different optical satellites	DOTA	GoogleEarth+	Pla+Ship+ST	0.1 ~ 4.5	800 × 800 ~ 4k × 4k	792/6384	89875
	xView	WorldView-3	Pla+Ship+ST	0.3	2.5k × 2.5k ~ 5k × 3k	216/1878	11005
between satellite and UAV	UCAS-AOD	GoogleEarth	Car	—	1280 × 659	510	7115
	CARPK	UAV	Car	0.01 ~ 0.04	1280 × 720	1448	89774
from optical to SAR images	HRRSD	GoogleEarth	Ship	0.15 ~ 1.2	1k × 1k ~ 2k × 2k	2165	3975
	SSDD	RadarSat-2+	Ship	1 ~ 15	300 × 300 ~ 500 × 500	1160	2587

- GoogleEarth+: GoogleEarth, Satellite JL-1, GF-2
- RadarSat-2+: RadarSat-2+, TerraSAR-X, Sentinel-1
- Pla+Ship+ST: Plane, Ship, Storage-tank
- The number of images for DOTA and xView is given as "original/cropped"

E. Network Training

The proposed VLSDA framework is trained through a two-stage process under a teacher–student paradigm. The vision–language encoder is kept frozen throughout training. Only the student detector, projection layers (used to map DINO features into the VLM space), and domain discriminators are updated.

1) Two-Stage Training Strategy: Training proceeds through the warm-up stage and self-training stage. The distinction between stages lies in whether pseudo-labels from the target domain are utilized.

During the warm-up stage, only labeled source-domain data are used for detection loss supervision. However, both source and target images are fed into the detector to extract features for alignment. Since no pseudo-labels are available for the target domain, object crops for VLM prototype extraction are only generated from source ground-truth boxes. As a result, the VLPA and GCCA modules operate using prototypes extracted solely from the source domain and VLM at this stage. This allows the model to establish stable initial feature alignment under clean supervision, mitigating early stage instability.

During the subsequent self-training stage, the teacher model generates pseudo-labels for the unlabeled target domain. These pseudo-labels serve dual roles: 1) they provide additional detection supervision for the student model and 2) they supply object crops to construct target-domain VLM prototypes. VLPA and GCCA modules are then applied to prototypes derived jointly from both source and target domains, progressively refining cross-domain alignment.

2) Detection Supervision: Detection is supervised by a set-based Hungarian matching loss following DINO [39]. For source-domain images, ground-truth labels are used directly. For the target domain, pseudo-labels with confidence above a threshold are selected for training. We adopt the confidence thresholding strategy following [18] to reduce label noise. The overall detection loss is as follows:

$$\mathcal{L}_{\text{det}} = \mathcal{L}^{s}_{\text{det}} + \mathcal{L}^{t}_{\text{det}}. \tag{13}$$

3) Total Training Loss: VLPA performs tridomain adversarial alignment over category prototypes, while GCCA enforces global-supervised contrastive alignment across classes. We further apply a feature-level domain adversarial loss $\mathcal{L}_{\text{backbone}}$ on multiscale backbone features following prior work:

$$\mathcal{L}_{\text{backbone}} = -\sum_{l} \left(\log D_l\left(\mathbf{F}^{s}_{l}\right) + \log\left(1 - D_l\left(\mathbf{F}^{t}_{l}\right)\right)\right) \tag{14}$$

where D_l denotes the domain discriminator for the lth feature scale, and GRL is applied before each discriminator to enable adversarial training. The complete training loss is expressed as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}} + \lambda_{\text{backbone}}\mathcal{L}_{\text{backbone}} + \lambda_{\text{VLPA}}\mathcal{L}_{\text{VLPA}} + \lambda_{\text{GCCA}}\mathcal{L}_{\text{GCCA}}. \tag{15}$$

The weight coefficients $\lambda_{\text{backbone}}$, λ_{VLPA}, and λ_{GCCA} are tuned empirically, and their choices will be analyzed in Section IV.

4) Teacher Model Update: The teacher model shares the same architecture as the student and is updated by the exponential moving average (EMA) of the student's weights at each iteration

$$\theta^{t}_{T} \leftarrow \beta\theta^{t-1}_{T} + (1-\beta)\,\theta^{t}_{S} \tag{16}$$

with $\beta = 0.999$ in all experiments, following prior works.

IV. EXPERIMENTS

A. Datasets

In this study, we conduct comprehensive evaluations under three widely adopted domain adaptation scenarios in remote sensing, following the experimental settings of Remote Sensing Teacher [18]. Six publicly available remote sensing datasets are employed to examine the effectiveness and generalization capability of the proposed approach across diverse cross-domain conditions. Specifically, we explore three types of domain shifts.

Cross-satellite adaptation evaluating knowledge transfer between satellite platforms, with xView [43] as the source domain and DOTA [44] as the target domain, following the setting of [17].

Satellite–unmanned aerial vehicle (UAV) adaptation assessing generalization across aerial platforms, using UCAS-AOD [45] ↔ CARPK [46].

Cross-modality adaptation examining transferability from optical to SAR imagery using HRRSD [47] as the source and SSDD [48] as the target, as suggested in [49].

An overview of the datasets and domain settings is provided in Table I, and detailed descriptions are presented in the following.

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1) *DOTA:* The DOTA dataset is one of the largest publicly available optical remote sensing datasets, developed by Wuhan University. It consists of 2806 aerial images collected from Google Earth, annotated with 188 282 object instances across 15 categories, including airplane, ship, and vehicle. To align with the xView dataset, we select three common categories (airplane, ship, and storage tank) for evaluation. Images are cropped into 800×800 patches with a stride of 100 pixels, and the model is trained and validated using the official trainval splits.
2) *xView:* Captured by the WorldView-3 satellite, xView is a high-resolution satellite image dataset containing over 1 million bounding boxes from various global scenes. Due to its fine-grained annotations, we merge subcategories into broader categories to match DOTA's taxonomy (e.g., motorboat, sailboat, and fishing vessel are merged into the ship category). The same three categories—airplane, ship, and storage tank—are used in our experiments. Images are similarly cropped to 800×800 with 100-pixel strides, and training follows the official splits.
3) *UCAS-AOD:* Released in 2014 by the University of Chinese Academy of Sciences, UCAS-AOD contains remote sensing images from Google Earth, designed with a uniform distribution of object orientations. In our experiments, we focus on the car category, comprising 510 images and 7114 annotated vehicles. Images are of size 1280×659, and no cropping is required. As there is no official split, we adopt a reproducible partitioning strategy: images with filenames ending in digits 1, 5, or 9 form the testing set, while the remaining are used for training (maintaining a 7:3 training–testing ratio).
4) *CARPK:* The CARPK dataset is collected by UAVs over parking lots, and each 1280×720 image contains only vehicles from a single class (car). This dataset complements UCAS-AOD for evaluating domain adaptation between satellite and UAV imagery. All images are used in their original resolution without further cropping.
5) *HRRSD:* The HRRSD dataset, released by UCAS in 2019, includes thousands of images sourced from Google Earth and Baidu Maps with resolutions ranging from 0.15 to 1.2 m. We select ship-related samples, resulting in 2165 images and 3975 instances. Input images are resized such that the shorter edge is at least 600 pixels. All samples are used for training.
6) *SSDD:* The SSDD contains 1160 SAR images with 2456 annotated ships. The dataset encompasses diverse maritime conditions and sensor variations. Following prior work [49], we pair SSDD with HRRSD to evaluate cross-modal transfer from optical to SAR imagery. Input images are resized to a minimum short edge of 600 pixels, and training/testing follows the official dataset split.

B. Evaluation Metric

To quantitatively assess the performance of our proposed method in DAOD, we adopt the mean average precision (AP) at an intersection-over-union (IoU) threshold of 0.5 (mAP_{50}) as the primary evaluation metric. This metric is widely used in remote sensing and object detection tasks due to its interpretability and consistency with prior works [17], [18].

Given a set of predicted bounding boxes and ground-truth annotations, the precision–recall curve is computed for each object category. The area under the precision–recall curve is defined as the AP for that category. The mean AP is then calculated by averaging the AP values across all evaluated categories.

C. Implementation Details

We adopt DINO as the base detector throughout all experiments. All models are trained on a single NVIDIA RTX 4090 GPU with 24-GB memory. Mixed-precision training is enabled to improve training efficiency and reduce GPU memory consumption.

Following common practice, ResNet-50 pretrained on ImageNet [50] is used as the backbone whenever applicable for both our method and all comparison baselines to ensure fairness. The training schedule consists of two stages as described in Section III-E: a warm-up stage and a self-training stage. The warm-up stage runs for 36 epochs, followed by ten self-training epochs, resulting in a total of 46 epochs per experiment.

In both stages, the Adam optimizer [51] is used with an initial learning rate of 1×10^{-4} and $\beta_1 = 0.9$. The learning rate is decayed by a factor of 0.1 at the 30th epoch. The EMA momentum for teacher model updates is set to 0.999.

The hyperparameters for domain alignment losses, including VLPA, GCCA, and backbone alignment, are set as $\lambda_{\text{VLPA}} = 0.1$, $\lambda_{\text{GCCA}} = 0.01$, and $\lambda_{\text{backbone}} = 0.2$ by default. The selection and sensitivity of these loss weights will be further discussed in the ablation studies.

D. Comparison With the State-of-the-Art Methods

To comprehensively evaluate the effectiveness of our proposed VLSDA framework, we conduct experiments on three representative remote sensing domain adaptation scenarios that involve shifts across sensors, platforms, and modalities. We compare our method against a broad range of state-of-the-art DAOD approaches, including CNN-based methods and Transformer-based methods.

In particular, since Remote Sensing Teacher reports results on the same datasets used in our experiments, we directly cite their performance as reported in the original paper. For VLDadaptor, originally developed for natural image domain adaptation, we reimplement the method on remote sensing data using the official code and replace its default CLIP weights with our remote sensing-specific pretrained model to ensure a fair comparison. The same applies to DATR, which we retrain in our setting using its official configuration. For other methods, we follow the reported results from Remote Sensing Teacher [18] and FIE-Net [20], which serves as a standardized benchmark for remote sensing DAOD.

As shown in Tables II and III, we report both source-only and Oracle (target-only) results as performance bounds

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TABLE II
EXPERIMENTAL RESULTS (%) OF THE SCENARIO BETWEEN DIFFERENT OPTICAL SATELLITES: XVIEW → DOTA

Method	Type	Backbone	Plane	Ship	Storage-tank	mAP_{50}
Source Only Deformable-DETR(ICLR'21)[52]	Transformer	resnet-50	64.8	55.5	32.6	51.0
Oracle Deformable-DETR(ICLR'21)[52]			93.5	74.1	76.2	81.3
Source Only DINO(ICLR' 23)[39]	Transformer	resnet-50	59.9	38.8	53.5	50.7
Oracle DINO(ICLR' 23)[39]			94.0	78.5	74.4	82.3
DAF(CVPR'18)[53]	CNN	resnet-50	57.3	34.3	36.4	42.7
SWF(CVPR'19)[11]	CNN	resnet-50	48.1	33.9	34.3	38.7
EPM(ECCV'20)[54]	CNN	resnet-101	60.5	59.8	43.9	54.8
SFA(ACM MM'21)[28]	Transformer	resnet-50	65.2	57.1	38.9	53.7
AQT(IJCAI'22)[29]	Transformer	resnet-50	68.2	<u>59.9</u>	46.8	58.3
FADA(TGRS'22)[17]	CNN	resnet-50	68.0	43.4	46.2	52.5
Remote Sensing Teacher(TGRS'24)[18]	Transformer	resnet-50	<u>75.7</u>	59.7	54.7	63.3
DATR(TIP'25)[19]	Transformer	resnet-50	70.9	54.9	**65.3**	<u>63.7</u>
FIE-Net(TGRS'25)[20]	CNN	vgg-16	66.8	52.6	45.9	55.1
VLDadaptor(TMM' 24)[6]	CNN	resnet-50	60.5	57.7	53.9	57.4
VLSDA	Transformer	resnet-50	**78.3**	**61.6**	<u>64.8</u>	**68.3**

- Highest result is shown in bold, while second best is underlined.

TABLE III
EXPERIMENTAL RESULTS (%) OF THE SCENARIO SATELLITE TO UAV: UCAS-AOD → CARPK, UAV TO SATELLITE: CARPK → UCAS-AOD, AND OPTICAL TO SAR IMAGES: HRRSD → SSDD

Method	Type	Backbone	UCAS-AOD → CARPK	CARPK → UCAS-AOD	HRRSD → SSDD
Source Only Deformable-DETR(ICLR'21)[52]	Transformer	resnet-50	61.2	58.1	27.8
Oracle Deformable-DETR(ICLR'21)[52]			80.9	79.1	93.6
Source Only DINO(ICLR' 23)[39]	Transformer	resnet-50	52.8	58.4	25.4
Oracle DINO(ICLR' 23)[39]			76.4	80.9	95.1
DAF(CVPR'18)[53]	CNN	resnet-50	68.4	53.7	47.0
SWF(CVPR'19)[11]	CNN	resnet-50	58.3	57.1	52.5
EPM(ECCV'20)[54]	CNN	resnet-101	71.2	65.0	54.9
SFA(ACM MM'21)[28]	Transformer	resnet-50	49.8	66.1	34.3
AQT(IJCAI'22)[29]	Transformer	resnet-50	71.0	70.8	39.9
Remote Sensing Teacher(TGRS'24)[18]	Transformer	resnet-50	76.2	75.6	58.5
DATR(TIP'25)[19]	CNN	resnet-50	76.4	73.1	49.5
FIE-Net(TGRS'25)[20]	CNN	vgg-16	<u>79.4</u>	<u>78.9</u>	58.5
VLDadaptor(TMM' 24)[6]	CNN	resnet-50	74.4	63.9	<u>60.3</u>
VLSDA	Transformer	resnet-50	**79.8**	**79.9**	**70.5**

under each scenario. Source-only refers to training solely on labeled source-domain data without any adaptation, while Oracle indicates full supervision on the target domain. These two settings provide lower and upper bounds for evaluating the effectiveness of domain adaptation methods.

Across all scenarios, DINO and Deformable-DETR exhibit comparable performance. In particular, Deformable-DETR tends to outperform DINO in settings with dense small objects (e.g., UCAS-AOD → CARPK), likely due to its superior local refinement capabilities. It is important to note that our choice of DINO as the base detector follows the configuration of DATR and does not confer significant performance advantages over Deformable-DETR in the context of remote sensing domain adaptation. This ensures that performance gains observed in our method are attributable to the proposed VLSDA framework, rather than differences in detector architecture.

1) Adaptation Between Different Optical Satellites: This scenario involves substantial domain shifts due to different satellite sensors, resulting in variations in spatial resolution, viewing angles, and annotation quality. As shown in Table II, our VLSDA method achieves a mAP_{50} of 68.3%, clearly outperforming previous state-of-the-art methods, including DATR and Remote Sensing Teacher. The most significant gains are

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TABLE IV
ABLATION STUDY OF VLSDA

VLPA	GCCA	mAP_{50}
×	×	64.2
×	√	66.7
√	×	66.8
√	√	**68.3**

observed in categories such as ship and plane, where visual ambiguity and background clutter often hinder adaptation. We note that for the storage-tank category, our result ranks second, slightly below DATR. This class exhibits less distinctive semantic cues compared with visually salient categories such as ship or plane, making it harder for the VLM to form robust language-grounded prototypes. Nevertheless, the overall mAP demonstrates that our triangular alignment achieves stronger generalization across heterogeneous optical domains.

2) Adaptation Between Satellite and UAV: This setting presents a severe cross-platform challenge due to differences in imaging altitude, object scale, and viewpoint (top down versus oblique). In the UCAS-AOD → CARPK transfer, VLSDA achieves 79.8%, outperforming DATR, VLDadaptor, and even the Oracle DINO, which is trained with full supervision on the target domain. This indicates that the background-aware hard negative sampling and vision–language alignment in VLSDA provide strong guidance even beyond what the supervised model learns. In the reverse direction, CARPK → UCAS-AOD, VLSDA again leads with 79.9%, confirming its robustness under significant geometric and density differences.

3) Adaptation From Optical to SAR Images: Among the three scenarios, HRRSD → SSDD represents the most challenging case due to the modality gap between optical and SAR imagery. Here, low-level texture features vary drastically, and many standard alignment techniques struggle. VLSDA achieves a clear performance lead, reaching a mAP_{50} of 70.5%, which surpasses the previous best by + 10.2 percentage points. Notably, it also exceeds the DATR method and Remote Sensing Teacher by a large margin. This demonstrates the strength of our vision–language-guided contrastive alignment in bridging cross-modal gaps where traditional low-level features fail. Particularly in challenging SAR scenes with high structural noise and semantic ambiguity, VLSDA maintains stable alignment, benefiting from both hard negatives and external VLM priors.

E. Ablation Studies

We conduct extensive ablation studies on the xView → DOTA scenario to systematically analyze the effectiveness of each proposed module in our VLSDA framework and to investigate the sensitivity of our method with respect to critical hyperparameters.

1) Ablation Each Component In VLSDA: Table IV evaluates the impact of our proposed modules, namely, VLPA and GCCA. We set a baseline that does not include any of the modules (i.e., only use backbone alignment), which achieves a mAP_{50} of 64.2%. Adding GCCA alone significantly increases performance to 66.7%, demonstrating that the contrastive alignment strategy effectively improves semantic consistency across domains. Likewise, adding VLPA alone yields a comparable gain, achieving a mAP_{50} of 66.8%, highlighting the effectiveness of our tridomain prototypical alignment scheme. Finally, combining both VLPA and GCCA achieves the best performance (68.3%), clearly demonstrating that these two modules are complementary and jointly enhance the model's adaptation capabilities.

TABLE V
ABLATION STUDY OF HYPERPARAMETER

$\lambda_{backbone}$	λ_{VLPA}	λ_{GCCA}	mAP_{50}
0.02	0.1	0.0025	64.6
2	0.1	0.0025	65.4
0.2	0.01	0.0025	65.8
0.2	1	0.0025	64.0
0.2	1	0.025	65.1
0.2	1	0.00025	65.6
0.2	0.1	0.0025	**68.3**

TABLE VI
ABLATION STUDY OF TEMPERATURE

τ	mAP_{50}
0.01	65.4
0.03	66.4
0.05	**68.3**
0.07	66.6
0.09	65.6

2) Ablation Hyperparameter: We further investigate the robustness of our method to key hyperparameters, including $\lambda_{backbone}$, λ_{VLPA}, and λ_{GCCA}. As shown in Table V, the optimal parameter combination achieves the highest mAP_{50}. Deviating from this optimal setting typically leads to performance degradation. Specifically, overly small or large backbone-alignment coefficients result in clear drops in accuracy. Similarly, significantly altering the values of λ_{VLPA} and λ_{GCCA} from their optimal settings also negatively impacts performance, confirming the importance of carefully balancing the alignment strengths across modules. Nonetheless, performance fluctuations remain within acceptable ranges (±3% to 4%), highlighting the relative robustness of VLSDA to minor hyperparameter variations.

We evaluate the influence of the temperature parameter in the GCCA contrastive loss. As shown in Table VI, performance varies nonmonotonically with temperature. A too-small value (0.01) reduces the emphasis on hard negatives, leading to less discriminative feature separation and a mAP_{50} drop to 65.4%. Conversely, a too-large value (0.09) overemphasizes a few hardest negatives, which can cause instability and degrade generalization, resulting in 65.6% mAP_{50}. The best result

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TABLE VII
ABLATION STUDY OF VLPA

$\mathcal{D}_{ST}$	$\mathcal{D}_{SV}$	$\mathcal{D}_{TV}$	mAP_{50}
×	×	×	64.2
×	×	√	65.6
×	√	×	63.7
×	√	√	66.0
√	×	×	63.1
√	×	√	64.6
√	√	×	64.8
√	√	√	**66.8**

(68.3%) is obtained at 0.05, which provides a balanced gradient scale between positive attraction and negative repulsion. This moderate setting encourages stable optimization while maintaining strong interclass separability.

3) Ablation Each Component in the VLPA: To better isolate the effect of each adversarial branch in the proposed VLPA module, we disable the GCCA component and conduct a detailed ablation on the three discriminators: $\mathcal{D}_{\mathrm{ST}}, \mathcal{D}_{\mathrm{SV}}$, and $\mathcal{D}_{\mathrm{TV}}$, which correspond to source–target, source–VLM, and target–VLM alignment, respectively.

As shown in Table VII, the target–VLM alignment $\mathcal{D}_{\mathrm{TV}}$ achieves the best performance, while the source–target alignment $\mathcal{D}_{\mathrm{ST}}$ even underperforms the baseline. This verifies our hypothesis that noisy and unstable target representations—caused by large distribution shifts in DAOD—can mislead binary S–T alignment, whereas anchoring the target domain to the semantically stable VLM space alleviates this problem. Interestingly, the source–VLM branch $\mathcal{D}_{\mathrm{SV}}$ brings no improvement, which is reasonable because the source domain already has reliable supervision from its ground-truth labels, making additional VLM guidance redundant.

For pairwise configurations, the combination of $\mathcal{D}_{\mathrm{SV}} + \mathcal{D}_{\mathrm{TV}}$ performs the best. This result highlights a key insight of alignment symmetry: when both the source and target domains are pulled toward the same semantic reference (the VLM), the alignment forces are geometrically balanced around a stable center. In contrast, asymmetric combinations such as $\mathcal{D}_{\mathrm{ST}} + \mathcal{D}_{\mathrm{SV}}$ or $\mathcal{D}_{\mathrm{ST}} + \mathcal{D}_{\mathrm{TV}}$ bias the optimization toward one side (source- or target-dominant), leading to suboptimal convergence.

Finally, enabling all three branches establishes a complete tridomain symmetric structure, where S, T, and VLM act as mutually constrained vertices of a triangle. The resulting equilibrium encourages both feature stability (via the fixed VLM domain) and discriminative consistency (via S–T alignment).

F. Effect of Vision–Language Backbones

To evaluate the robustness of our framework under different VLM backbones, we conduct experiments using four VLM variants drawn from two representative remote sensing VLMs, LRSCliP [55] and RS5M [33]. These backbones cover diverse image encoder scales and patch sizes, allowing us to assess the influence of semantic granularity and model capacity on cross-domain alignment. Specifically, ViT-B/16 and ViT-L/14 denote Vision Transformers of different sizes (base and large) and patch resolutions (16×16 and 14×14 pixels, respectively).

As shown in Table VIII, the model using RS5M ViT-B/32 achieves the best overall performance, particularly on the HRRSD → SSDD task. Interestingly, the larger ViT-L/14 backbone does not yield further improvements. This may stem from the substantial gap between its 768-D embeddings and the 256-D feature space of the DINO-based detector, making it harder to project both into a unified embedding space. This observation indicates that, even when VLMs serve as an additional and stable semantic domain, the compatibility between the VLM encoder and the detector backbone remains crucial for effective cross-domain supervision.

Moreover, VLMs with larger patch sizes tend to perform better in remote sensing scenarios. This can be attributed to the prevalence of small objects in remote sensing imagery, where image tiles are often upsampled to meet the input resolution requirements of ViT encoders. Larger patch sizes reduce the degree of forced upsampling, thereby mitigating the impact of interpolation artifacts and preserving the structural integrity of small objects.

G. Limitations and Discussion

The Vision Transformers adopted in our VLMs require fixed-size image inputs, which can be suboptimal for remote sensing imagery dominated by small objects. When image tiles are upsampled to meet the input resolution of the VLM encoder, excessive interpolation may distort spatial details and introduce noise into the extracted features. This limitation affects the fine-grained representation of small targets, especially when they occupy only a small portion of the image. A potential remedy would be to process full-size remote sensing images directly through the VLM and employ text-guided localization to extract region-level features. Such a design could enhance semantic grounding and yield more accurate anchors for cross-domain alignment. However, realizing this approach would require modifying or fine-tuning the VLM architecture, which would increase computational cost.

H. Visualization and Analysis

1) Detection Results: We present qualitative detection results of our VLSDA framework and compare them with the state-of-the-art methods (DATR and VLDadaptor), as well as the source-only and DINO Oracle baselines, across different adaptation scenarios. Representative examples are shown in Fig. 4.

As illustrated in xView to DOTA results, VLSDA achieves clearer and more accurate bounding-box predictions compared with other methods. For instance, in the ship detection scenario, VLSDA successfully identifies most small vessels and suppresses false detections, outperforming DATR and VLDadaptor, which generate considerable false positives or incomplete detections. This visual improvement aligns with the quantitative performance boost observed in Table II.

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TABLE VIII
EXPERIMENTAL RESULTS (mAP_{50}) OF VLSDA UNDER DIFFERENT VLMS

VLM Backbone	XVIEW → DOTA	UCAS-AOD → CARPK	CARPK → UCAS-AOD	HRRSD → SSDD
LRSCLIP ViT-B/16	66.1	78.9	79.0	64.7
LRSCLIP ViT-L/14	64.6	**79.9**	78.7	64.7
RS5M ViT-B/32	**68.3**	79.8	**79.9**	**70.5**
RS5M ViT-L/14	66.8	**79.9**	78.0	61.4

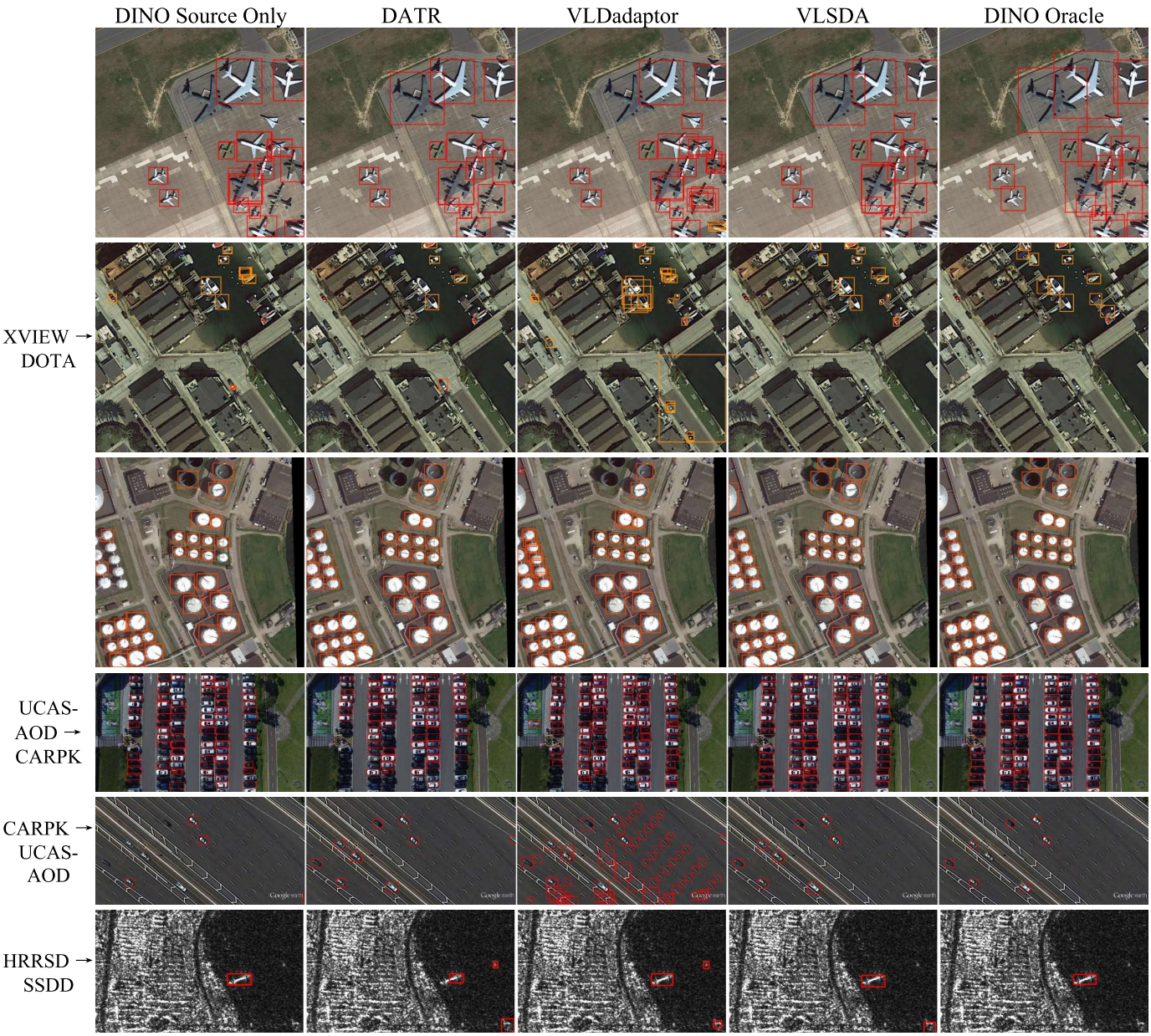


Fig. 4. Our detection results compared with the state-of-the-art methods. Different categories are marked with different colors. The confidence threshold for visualization is set to 0.3. "Source-only" represents the base detector that uses only source-domain data for training. Other descriptions represent their respective methods.

The UCAS-AOD to CARPK and CARPK to UCAS-AOD depict dense car detection under significant viewpoint and scale differences. Our VLSDA method provides more precise localization and fewer missed detections compared with both source-only and other state-of-the-art methods. VLSDA closely matches Oracle quality, achieving precise alignment and reducing missed or duplicated detections, emphasizing the effectiveness of our VLM-guided alignment strategy.

The last HRRSD to SSDD demonstrates adaptation from optical to SAR images, arguably the most challenging setting

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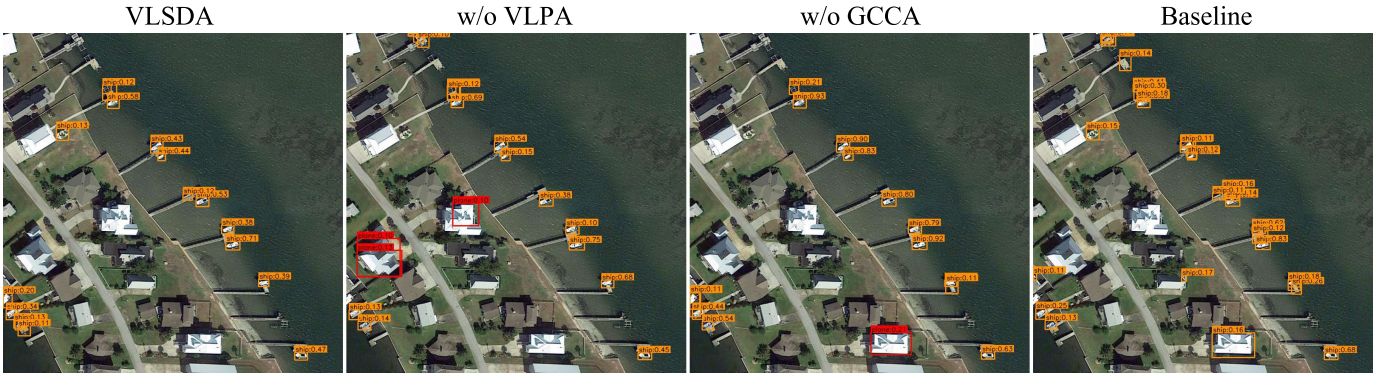


Fig. 5. Qualitative ablation results on the xView → DOTA scenario. Different categories are marked with different colors.

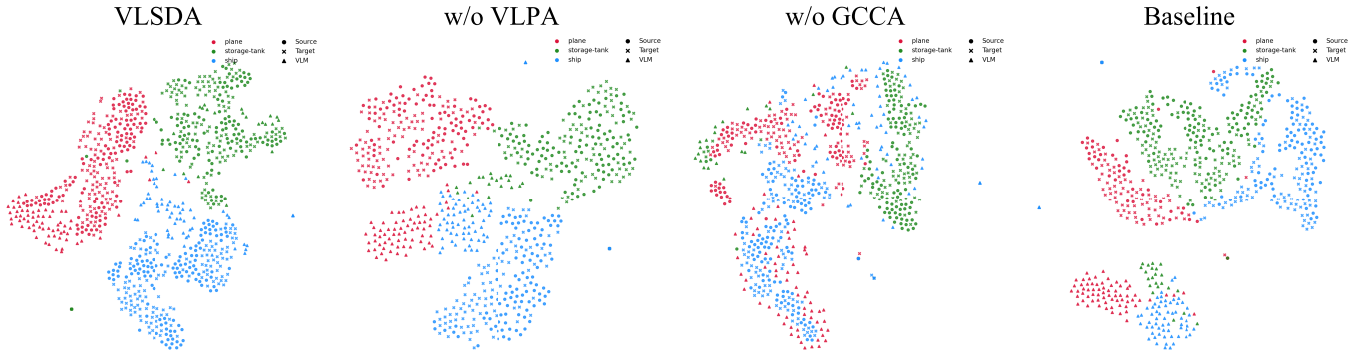


Fig. 6. t-SNE visualization of object prototypes across domains (xView → DOTA). Colors denote object categories, while markers indicate domains (●: Source, ×: Target, and ▲: VLM).

due to large modality gaps and significant image distortion. VLSDA maintains robust detection quality, correctly localizing ships under severe SAR noise, whereas other methods either fail to detect or produce false alarms. The stable detection accuracy by VLSDA, comparable to Oracle performance, illustrates the advantage of contrastive learning with external semantic supervision in bridging low-level modality discrepancies.

2) Ablation Results: To further evaluate the contribution of each component, Fig. 5 visualizes the detection results under different ablation settings. The complete VLSDA model yields the most accurate and comprehensive detections with minimal false alarms. When removing VLPA (w/o VLPA) or GCCA (w/o GCCA), feature alignment becomes unstable, causing objects such as buildings to be misclassified as ships or planes due to weakened intraclass consistency. The degradation is further amplified when both modules are removed (Baseline), leading to many misplaced predictions.

3) Feature Visualization: To further illustrate the effect of each component, we visualize the learned features on the xView → DOTA scenario using t-SNE [56]. As shown in Fig. 6, different colors represent object categories (plane, storage tank, and ship), while different markers denote the source (●), target (×), and VLM (▲) domains.

In our VLSDA model, the features from source, target, and VLM domains are well aligned within each class cluster, while distinct categories remain clearly separable in the embedding space. When VLPA is removed, the VLM prototypes drift away from the detector clusters, showing that the alignment between detection features and VLM embeddings becomes unstable. In addition, removing GCCA causes severe interclass entanglement: features of different categories overlap in the shared space, showing that GCCA plays a crucial role in maintaining discriminative class structure. Finally, when both VLPA and GCCA are removed, the gap between VLM and detector features becomes the largest, and the class clusters of source and target domains are no longer clearly separated. These observations validate that VLPA ensures "where to align," while GCCA ensures "how to separate."

V. Conclusion

We present VLSDA, a domain adaptation framework supervised by a frozen VLM image encoder that serves as an additional and stable semantic domain. By treating the VLM as an equal third domain and performing tridomain (source–target–VLM) alignment, VLSDA guides cross-domain learning toward external semantic anchors, thereby reducing reliance on unstable target-domain representations that commonly arise in remote sensing.

To achieve robust and discriminative adaptation, VLSDA integrates two complementary modules in a unified loop. VLPA employs a tridomain adversarial alignment that aggregates source and target features toward VLM-derived semantic prototypes, while GCCA reinforces class-aware discriminability through global contrastive constraints. Acting jointly, they

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guide features toward domain-agnostic anchors and maintain clear interclass separation, effectively enhancing training robustness.

Extensive experiments across three representative adaptation scenarios, including satellite to satellite, satellite to UAV, and optical to SAR, demonstrate that VLSDA consistently outperforms existing state-of-the-art methods. In particular, VLSDA attains 70.5% mAP_{50} in the optical to SAR setting, evidencing strong resilience to large modality gaps. These results indicate that integrating external semantic anchors is an effective and scalable strategy for DAOD in remote sensing.

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